

# RichnessSelection: math and numerical recipe for optical cluster selection bias

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The mean lensing surface density around optically selected clusters of observed richness  $\lambda^{\text{ob}}$  at observed redshift  $z^{\text{ob}}$  is the sum of a one-halo term (the cluster’s own miscentered NFW profile) and a two-halo term (correlated neighbouring halos along the line of sight):

$$\langle \Sigma(R \mid \lambda^{\text{ob}}, z^{\text{ob}}) \rangle = \langle \Sigma^{\text{1h}}(R \mid \lambda^{\text{ob}}, z^{\text{ob}}) \rangle + \langle \Sigma^{\text{prj}}(R \mid \lambda^{\text{ob}}, z^{\text{ob}}) \rangle, \quad (1)$$

where  $\langle \Sigma^{\text{1h}} \rangle$  is evaluated from the miscentered NFW of the target halo and  $\langle \Sigma^{\text{prj}} \rangle$  is the projected surface density that **RichnessSelection** implements. The physical origin of the selection effect is that REDMAPPER counts richness through red-sequence member galaxies, and red galaxies are biased tracers of large-scale structure: their correlation with the surrounding density field means that, at fixed halo mass, a line of sight with more red galaxies gets assigned a larger  $\lambda^{\text{ob}}$  than one with fewer. The two-halo term at fixed  $\lambda^{\text{ob}}$  therefore does not carry the bare halo bias  $b(M, z)$ ; Costanzi et al. [2026] show it must instead carry a *selection-affected* cluster bias  $b_{\text{sel}}(\lambda^{\text{ob}}, z^{\text{ob}}, \theta)$  that interpolates between a small-scale limit (dominated by close red-galaxy neighbours,  $b_{\text{sel}}^{\text{small}}$ ) and a large-scale limit (set by correlated LSS at the red-galaxy population bias,  $b_{\text{sel}}^{\text{large}}$ ). Concretely, rather than writing the two-halo term with the halo-bias aggregate  $b_{\text{halo}}$  as in a standard lensing pipeline, we factor the scale-dependent selection effect into  $b_{\text{sel}}$ ,

$$\langle \Sigma^{\text{prj}}(R \mid \lambda^{\text{ob}}, z^{\text{ob}}) \rangle = \int d\theta \sin \theta \int dz \frac{dV}{dz d\Omega} \int dM n(M, z) [1 + b(M, z) b_{\text{sel}}(\lambda^{\text{ob}}, z^{\text{ob}}, \theta) \xi_{\text{NL}}] \Sigma_{\text{mis}}, \quad (2)$$

(the full form with all arguments is eq. (14)). The package returns  $b_{\text{sel}}$  and the resulting  $\langle \Sigma^{\text{prj}}(R \mid \lambda^{\text{ob}}, z^{\text{ob}}) \rangle$  at the  $(\lambda^{\text{ob}}, z^{\text{ob}})$  of a DES Y3 cluster bin in  $\sim 30$  ms per call.

This note has three parts. The **model** section gives the mathematical pipeline that assembles  $b_{\text{sel}}$  non-circularly from halo and projection ingredients, and the resulting two-halo profile  $\langle \Sigma^{\text{prj}}(R \mid \lambda^{\text{ob}}, z^{\text{ob}}) \rangle$ . The **numerical pitfalls** section documents the two subtle hazards we hit when putting that pipeline on a grid — the  $\theta$ -exclusion step at the halo-halo exclusion boundary and the photo- $z$  band structure of the  $z$ -integrand — and the GL-on-physics-split fixes that drive the measured error below 0.01% vs `scipy.integrate.quad`. The **API and glossary** section is a pointer table from each mathematical symbol to the Python call that returns it.

## Model

### 1 Scope and notation

`RichnessSelection` computes the two-halo projected profile  $\langle \Sigma^{\text{prj}}(R \mid \lambda^{\text{ob}}, z^{\text{ob}}) \rangle$  (eq. 14) together with the selection bias  $b_{\text{sel}}$  that feeds it, at one  $(\lambda^{\text{ob}}, z^{\text{ob}})$  per call. Photo- $z$  on the target, target miscentering, triaxiality, and REDMAPPER percolation member-loss are left out (only the  $\lambda < \lambda^{\text{ob}}$  ranking cut is kept). Linear  $P(k, z)$  from CAMB feeds  $\xi_{\text{NL}}$  via haloFit + FFTlog.

Masses are in  $h^{-1} M_{\odot}$ ;  $R$  is a physical projected distance;  $\theta$  is a sky angle;  $\chi, R_{\lambda}, R_{\text{excl}}$  are comoving in  $h^{-1}$  Mpc. Default cosmology is Buzzard-v1.1 ( $\Omega_m = 0.286$ ,  $h = 0.7$ ,  $\sigma_8 = 0.82$ ). Key shorthand:  $R_{\lambda}(\lambda^{\text{ob}}) = (\lambda^{\text{ob}}/100)^{0.2} h^{-1}$  Mpc,  $R_{\text{excl}} \equiv R_{\lambda}(\lambda^{\text{ob}})(1 + z^{\text{ob}})$ , and  $b_{\text{sel}}^{\text{small}}/b_{\text{sel}}^{\text{large}}$  are the two asymptotes of the sigmoid ansatz below.

### 2 Core equations

#### 2.1 Halo-bias aggregate $b_{\text{halo}}(\lambda^{\text{ob}}, z^{\text{ob}})$

A single 1-D  $\ln M$  integral at fixed  $z^{\text{ob}}$ , weighting the Tinker-2010 halo bias by the halo-mass distribution at fixed observed richness:

$$b_{\text{halo}}(\lambda^{\text{ob}}, z^{\text{ob}}) = \frac{\int d \ln M \, M \, n(M, z^{\text{ob}}) \, b(M, z^{\text{ob}}) \, P(\lambda^{\text{ob}} \mid M, z^{\text{ob}})}{\int d \ln M \, M \, n(M, z^{\text{ob}}) \, P(\lambda^{\text{ob}} \mid M, z^{\text{ob}})}. \quad (3)$$

Here  $M$  is the halo mass in  $h^{-1} M_{\odot}$ ;  $\ln M$  runs over the integration support  $[10^{13}, 10^{15.5}] h^{-1} M_{\odot}$ ;  $n(M, z)$  is the Tinker-2008 halo mass function (§6);  $b(M, z)$  is the Tinker-2010 peak-height halo bias; and  $P(\lambda^{\text{ob}} \mid M, z)$  is the mass–observable relation pdf from MOR. The factor of  $M$  inside the integrand converts  $dM$  to  $d \ln M$ .

#### 2.2 Sigmoid ansatz for $b_{\text{sel}}(\theta)$

Costanzi et al. [2026], eq. `b_sel_theta`:

$$b_{\text{sel}}(\lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}}, \theta) = b_{\text{sel}}^{\text{small}} [1 - \sigma(\theta)] + b_{\text{sel}}^{\text{large}} \sigma(\theta), \quad \sigma(\theta) = \frac{1}{1 + e^{-k(\theta - \theta_0)}}. \quad (4)$$

Here  $\theta$  is the sky angular separation from the target;  $b_{\text{sel}}^{\text{small}}$  and  $b_{\text{sel}}^{\text{large}}$  are the two plateaus (small- and large-angle limits of the bias, both scalars at fixed  $(\lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}})$ );  $\sigma(\theta)$  is the logistic transition function whose shape constants are fixed to  $k = 2.5/\theta_{\lambda}$  and  $\theta_0 = \theta_{\lambda}/2$ , with  $\theta_{\lambda} \equiv R_{\lambda}(\lambda^{\text{ob}})/D_A(z^{\text{ob}})$  the angle subtended by the target’s REDMAPPER radius  $R_{\lambda}(\lambda^{\text{ob}}) = (\lambda^{\text{ob}}/100)^{0.2} h^{-1}$  Mpc. The  $\theta$ -shape is fixed analytically; only the two plateaus  $b_{\text{sel}}^{\text{small}}$  and  $b_{\text{sel}}^{\text{large}}$  need to be computed.

#### 2.3 Projection kernel and $\mathcal{P}[X]$ operator

The *projection kernel*  $\rho^{\text{prj}}$  is the per-halo weight with which a projected halo at  $(\lambda, z, \theta)$  contaminates the observed richness of a target at  $(\lambda^{\text{ob}}, z^{\text{ob}})$  (Costanzi et al. 2026, eq. `weight`):

$$\rho^{\text{prj}}(\lambda, z, \theta \mid \lambda^{\text{ob}}, z^{\text{ob}}) = w_z(z, z^{\text{ob}}) f_A(\theta, \lambda^{\text{ob}}, z^{\text{ob}}, \lambda, z) \lambda \mathbf{1}_{\{\lambda < \lambda^{\text{ob}}\}}. \quad (5)$$

Here  $\lambda$  is the projected halo’s true richness;  $z$  is its redshift;  $\theta$  is its angular separation from the target;  $w_z(z, z^{\text{ob}})$  is the parabolic photo- $z$  kernel quantifying the probability that a galaxy at redshift

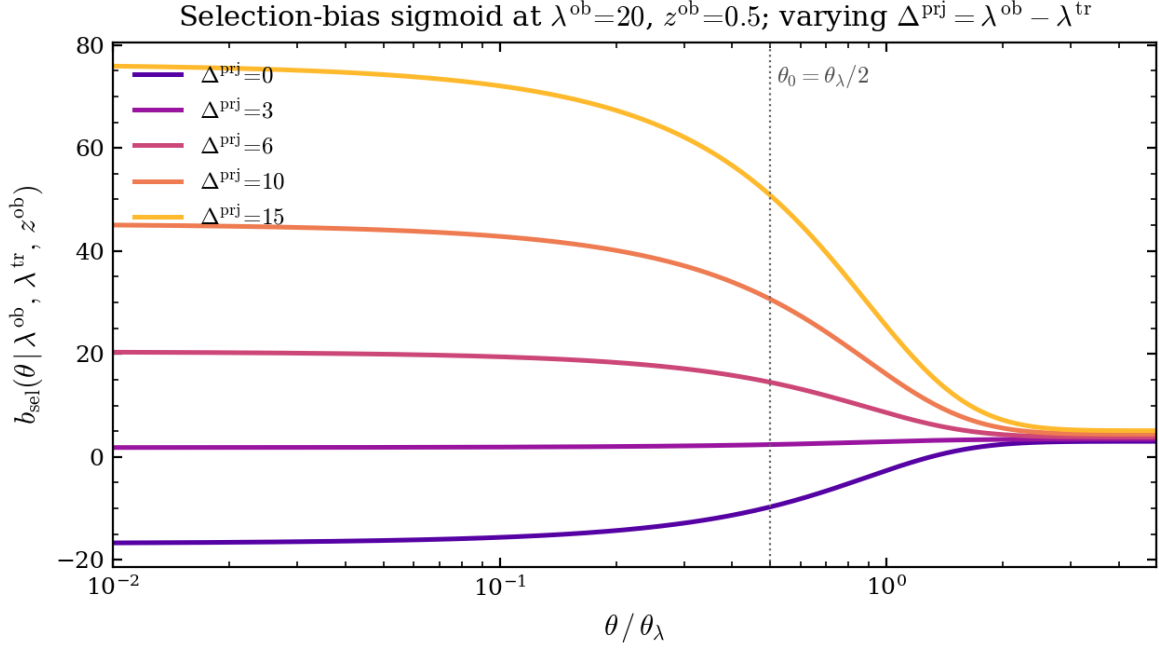


Figure 1: Selection-bias sigmoid  $b_{\text{sel}}(\theta | \lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}})$  at fixed  $(\lambda^{\text{ob}}, z^{\text{ob}}) = (20, 0.5)$  as we vary the projection contamination  $\Delta^{\text{prj}} \equiv \lambda^{\text{ob}} - \lambda^{\text{tr}}$ . The  $x$ -axis is scaled by the per-bin angle  $\theta_\lambda$ , so the sigmoid pivot  $\theta_0 = \theta_\lambda/2$  (dotted) is universal. The  $\theta$ -shape is fixed by the ansatz;  $\Delta^{\text{prj}}$  drives both plateaus through Steps 4–5. The left plateau  $b_{\text{sel}}^{\text{small}}$  (close neighbours projecting into the target aperture) grows fastest with  $\Delta^{\text{prj}}$ ; the right plateau  $b_{\text{sel}}^{\text{large}}$  (correlated large-scale structure) moves less because it is closed by the empirical  $b_{\text{halo}}[1 + 0.13 \delta^{\text{prj}}]$  relation.

$z$  is counted in the target’s aperture at  $z^{\text{ob}}$ ;  $f_A(\theta, \lambda^{\text{ob}}, z^{\text{ob}}, \lambda, z) \in [0, 1]$  is the closed-form overlap fraction of the two REDMAPPER disks (of the target and the projected halo); the factor  $\lambda$  converts “this halo is present” into “this halo contributes  $\lambda$  counts”; and the indicator  $\mathbf{1}_{\{\lambda < \lambda^{\text{ob}}\}}$  implements the REDMAPPER ranking cut (only projected halos of lower richness than the target contaminate the target’s aperture).

The  $\mathcal{P}[X]$  operator is the line-of-sight-averaged integral of an arbitrary  $X(M, z, \theta)$  weighted by the halo distribution and  $\rho^{\text{prj}}$ :

$$\begin{aligned} \mathcal{P}[X](\lambda^{\text{ob}}, z^{\text{ob}}) &\equiv \int dz \frac{dV}{dz d\Omega}(z) \int dM n(M, z) \int d\lambda P(\lambda | M, z) \\ &\cdot 2\pi \int d\theta \sin \theta \rho^{\text{prj}}(\lambda, z, \theta | \lambda^{\text{ob}}, z^{\text{ob}}) X(M, z, \theta | z^{\text{ob}}). \end{aligned} \quad (6)$$

The integrations marginalise, in order, over the projected halo’s redshift  $z$  (weighted by comoving volume per steradian  $dV/dz d\Omega$ ), mass  $M$  (weighted by the halo mass function  $n(M, z)$ ), true richness  $\lambda$  (weighted by the MOR  $P(\lambda | M, z)$ ), and angular separation  $\theta$  from the target (with the usual  $2\pi \sin \theta d\theta$  sky-element).

Three specialisations drive the pipeline:

$$\mathcal{P}[1] = \langle \Delta_{\text{bkg}}^{\text{prj}} \rangle(\lambda^{\text{ob}}, z^{\text{ob}}), \quad I_2 \equiv \mathcal{P}[b \xi_{\text{NL}}], \quad I_1 \equiv \mathcal{P}[b \xi_{\text{NL}} \sigma(\theta)], \quad (7)$$

where  $b = b(M, z)$  is the Tinker-2010 halo bias (the halo’s own clustering amplitude) and  $\xi_{\text{NL}} = \xi_{\text{NL}}(z, z^{\text{ob}}, \theta)$  is the nonlinear matter correlation function evaluated at the 3-D comoving separation  $\Delta\chi(z, z^{\text{ob}}, \theta)$  between the target at  $z^{\text{ob}}$  and the projected halo at  $(z, \theta)$ . By linearity,  $\mathcal{P}[b \xi_{\text{NL}}(1 - \sigma)] = I_2 - I_1$ . Halo exclusion sets  $\xi_{\text{NL}} \equiv 0$  whenever  $\Delta\chi < R_{\text{excl}}$ .

## 2.4 Bias pipeline

This subsection gives the calculation chain from halo and projection ingredients to the selection-affected cluster bias  $b_{\text{sel}}(\lambda^{\text{ob}}, z^{\text{ob}}, \theta)$  that eq. (14) requires. From the halo ingredients  $\{n(M, z), b(M, z), P(\lambda | M, z)\}$  and the projection ingredients  $\{w_z, f_A, \xi_{\text{NL}}\}$  it first evaluates the halo-bias aggregate  $b_{\text{halo}}(\lambda^{\text{ob}}, z^{\text{ob}})$  of eq. (3), a 1-D  $\ln M$  integral at fixed  $z^{\text{ob}}$ . In parallel, the hardest numerical block computes the three forward integrals  $\mathcal{P}[1]$ ,  $I_1$ ,  $I_2$  of eq. (6); the recipes for this step are the subject of §3–4 below. With these in hand, the random-aperture projection contamination

$$\Delta_{\text{RND}}^{\text{prj}}(\lambda^{\text{ob}}, z^{\text{ob}}) = \mathcal{P}[1](\lambda^{\text{ob}}, z^{\text{ob}}) + b_{\text{halo}}(\lambda^{\text{ob}}, z^{\text{ob}}) I_2(\lambda^{\text{ob}}, z^{\text{ob}}) \quad (8)$$

follows by substituting  $b_{\text{sel}} \rightarrow b_{\text{halo}}$  in the  $b_{\text{sel}}$ -linearised identity ( $b_{\text{halo}}$  pulls out of the mass integral because it is mass-independent). The large- $\theta$  plateau is then

$$b_{\text{sel}}^{\text{large}}(\lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}}) = b_{\text{halo}}(\lambda^{\text{ob}}, z^{\text{ob}}) [1 + 0.13 \delta^{\text{prj}}(\lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}})], \quad \delta^{\text{prj}} \equiv \frac{\lambda^{\text{ob}} - \lambda^{\text{tr}}}{\Delta_{\text{RND}}^{\text{prj}}} - 1, \quad (9)$$

where  $\delta^{\text{prj}}$  is the fractional excess contamination at  $(\lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}})$  relative to a random aperture and the empirical coefficient 0.13 is the Buzzard calibration. Finally the small- $\theta$  plateau closes the remaining linear equation in  $b_{\text{sel}}$ :

$$b_{\text{sel}}^{\text{small}}(\lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}}) = \frac{(\lambda^{\text{ob}} - \lambda^{\text{tr}}) - \mathcal{P}[1] - b_{\text{sel}}^{\text{large}} I_1}{I_2 - I_1}, \quad (10)$$

where the denominator  $I_2 - I_1 = \mathcal{P}[b \xi_{\text{NL}}(1 - \sigma)]$  is the contribution of the  $1 - \sigma$  (small- $\theta$ ) part of the sigmoid ansatz, and the numerator is what remains of the  $\Delta^{\text{prj}}$  identity once the  $\mathcal{P}[1]$  constant and the  $b_{\text{sel}}^{\text{large}}$ -weighted  $I_1$  piece have been subtracted. With  $b_{\text{sel}}^{\text{small}}$  and  $b_{\text{sel}}^{\text{large}}$  both known, eq. (4) returns  $b_{\text{sel}}(\lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}}, \theta)$ . Marginalising over  $\lambda^{\text{tr}}$  via

$$b_{\text{sel}}(\lambda^{\text{ob}}, z^{\text{ob}}, \theta) = \int d\lambda^{\text{tr}} b_{\text{sel}}(\lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}}, \theta) P(\lambda^{\text{tr}} | \lambda^{\text{ob}}, z^{\text{ob}}), \quad (11)$$

with  $P(\lambda^{\text{tr}} | \lambda^{\text{ob}}, z^{\text{ob}}) \propto P(\lambda^{\text{ob}} | \lambda^{\text{tr}}, z^{\text{ob}}) \pi(\lambda^{\text{tr}}, z^{\text{ob}})$  and  $\pi(\lambda^{\text{tr}}, z^{\text{ob}}) = \int dM n(M, z) P(\lambda^{\text{tr}} | M, z)$  the prior on the true richness, gives the final input to eq. (14).

Because eq. (4) is linear in the plateaus ( $b_{\text{sel}}^{\text{small}}, b_{\text{sel}}^{\text{large}}$ ) and the sigmoid  $\sigma(\theta)$  carries no  $\lambda^{\text{tr}}$  dependence, the  $\lambda^{\text{tr}}$ -marginalisation commutes with the sigmoid assembly. Defining the  $\lambda^{\text{tr}}$ -averaged plateaus

$$\bar{b}_{\text{small}}(\lambda^{\text{ob}}, z^{\text{ob}}) = \int d\lambda^{\text{tr}} b_{\text{sel}}^{\text{small}}(\lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}}) P(\lambda^{\text{tr}} | \lambda^{\text{ob}}, z^{\text{ob}}), \quad \bar{b}_{\text{large}}(\lambda^{\text{ob}}, z^{\text{ob}}) = \int d\lambda^{\text{tr}} b_{\text{sel}}^{\text{large}}(\lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}}) P(\lambda^{\text{tr}} | \lambda^{\text{ob}}, z^{\text{ob}}) \quad (12)$$

eq. (11) reduces to

$$b_{\text{sel}}(\lambda^{\text{ob}}, z^{\text{ob}}, \theta) = \bar{b}_{\text{small}}(\lambda^{\text{ob}}, z^{\text{ob}}) [1 - \sigma(\theta)] + \bar{b}_{\text{large}}(\lambda^{\text{ob}}, z^{\text{ob}}) \sigma(\theta). \quad (13)$$

Eq. (13) is numerically identical to eq. (11) (it is an algebraic rearrangement, not an approximation), but shifts the  $\lambda^{\text{tr}}$ -quadrature from *inside* the  $\theta$ -loop to a one-shot pre-computation at fixed  $(\lambda^{\text{ob}}, z^{\text{ob}})$ . Downstream consumers that assemble  $b_{\text{sel}}(\theta)$  at many  $\theta$ -nodes — notably  $\langle \Sigma^{\text{prj}} \rangle$  and  $\langle \Delta \Sigma^{\text{prj}} \rangle$  — therefore only pay the cost of the analytic sigmoid per call.

The  $b_{\text{sel}}^{\text{small}}$ -denominator guard in eq. (10) ( $|I_2 - I_1| \rightarrow 0 \Rightarrow b_{\text{sel}}^{\text{small}} \leftarrow b_{\text{sel}}^{\text{large}}$ ) is applied per- $\lambda^{\text{tr}}$  before the average, so the definition of  $\bar{b}_{\text{small}}$  in eq. (12) is well-defined even when the denominator is small for a subset of the  $\lambda^{\text{tr}}$  grid.

In `SelBias` (§4), steps computed once per  $(\lambda^{\text{ob}}, z^{\text{ob}})$  ( $b_{\text{halo}}$ ,  $\mathcal{P}[1]$ ,  $I_1$ ,  $I_2$ ,  $\Delta_{\text{RND}}^{\text{prj}}$ ) are cached inside `bias_precompute`; the per- $\lambda^{\text{tr}}$  closures of eqs. (9)–(10) live in `bias_from_precomp`; the sigmoid assembly at fixed  $\lambda^{\text{tr}}$  is `blob_theta`; and the  $\lambda^{\text{tr}}$ -marginalised plateaus ( $\bar{b}_{\text{small}}$ ,  $\bar{b}_{\text{large}}$ ) of eq. (12) are cached by `marginalised_plateaus`. The public entry point `b_sel_marginalised(theta, lob, zob)` evaluates eq. (13) directly from those cached plateaus.

## 2.5 Two-halo projected profile

Costanzi et al. [2026], eq. `sigprj`:

$$\langle \Sigma^{\text{prj}}(R \mid \lambda^{\text{ob}}, z^{\text{ob}}) \rangle = \int d\theta \sin \theta \int dz \frac{dV}{dz d\Omega} \int dM n(M, z) \cdot [1 + b(M, z) b_{\text{sel}}(\lambda^{\text{ob}}, z^{\text{ob}}, \theta) \xi_{\text{NL}}(z, z^{\text{ob}}, \theta)] \Sigma_{\text{mis}}(R \mid M, z, \theta, z^{\text{ob}}). \quad (14)$$

Here  $R$  is the physical projected distance around the target in  $h^{-1}$  Mpc;  $\theta$  is the angular separation of the contaminating halo from the target;  $z$  is the halo’s redshift;  $M$  its mass;  $n(M, z)$  the halo mass function;  $b(M, z)$  the halo bias;  $b_{\text{sel}}$  the selection-affected cluster bias from eq. (11);  $\xi_{\text{NL}}$  the nonlinear matter correlation function at the 3-D separation between the target and the  $(z, \theta)$  halo; and  $\Sigma_{\text{mis}}$  is the azimuth-averaged miscentered NFW surface density. The 1 inside the bracket is the mean (uncorrelated) contribution; the  $b b_{\text{sel}} \xi_{\text{NL}}$  term is the correlation-function boost that the two-halo term captures.

$\Sigma_{\text{mis}}$  is the one-halo surface density integrated over the azimuthal angle  $\varphi$  because the projected halo is off-centre with respect to the profile radius:

$$\Sigma_{\text{mis}}(R \mid M, z, \theta, z^{\text{ob}}) = \int_0^{2\pi} d\varphi \Sigma(R_h \mid M, z), \quad R_h = \sqrt{R^2 + R_\theta^2 - 2RR_\theta \cos \varphi}, \quad (15)$$

with  $R_\theta \equiv \theta D_A(z^{\text{ob}})$  the physical projected distance from the target to the contaminating halo and  $\Sigma(R_h \mid M, z)$  the centred NFW surface density of the contaminating halo at 2-D radius  $R_h$  from its own centre.  $\Sigma_{\text{mis}}$  is tabulated from the offset-NFW lookup (`nfw.py`), with a factor of 2 to match the  $\int_0^{2\pi}$  convention.

## 2.6 Lensing excess $\langle \Delta \Sigma^{\text{prj}}(R \mid \lambda^{\text{ob}}, z^{\text{ob}}) \rangle$

The two-halo observable returned by `DeltaSigmaPrj` is the radial excess of (14),

$$\langle \Delta \Sigma^{\text{prj}}(R \mid \lambda^{\text{ob}}, z^{\text{ob}}) \rangle \equiv \bar{\Sigma}^{\text{prj}}(< R \mid \lambda^{\text{ob}}, z^{\text{ob}}) - \langle \Sigma^{\text{prj}}(R \mid \lambda^{\text{ob}}, z^{\text{ob}}) \rangle. \quad (16)$$

Because  $R \rightarrow \bar{\Sigma}(< R) - \Sigma(R)$  is a linear functional in the radial argument only, it commutes with the  $(\theta, z, M)$  integrals in eq. (14): the pipeline of §3–4 carries over unchanged, and the only substitution is at the NFW lookup,  $\Sigma_{\text{mis}} \rightarrow \Delta \Sigma_{\text{mis}} \equiv \bar{\Sigma}_{\text{mis}}(< R) - \Sigma_{\text{mis}}(R)$ , evaluated via the companion `log_deltasigma_single` table shipped with the offset-NFW kit. The full derivation, the scale argument for adaptive  $\theta_{\text{max}}$ , and the factor-of-2 convention cross-check against the analytical Wright & Brainerd 2000 centred NFW  $\Delta \Sigma$  live in `docs/delta_sigma_prj_derivation.tex`.

**Profile vs projection contamination.** Figure 3 shows  $\langle \Delta \Sigma^{\text{prj}}(R \mid \lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}}) \rangle$  at  $(\lambda^{\text{ob}}, z^{\text{ob}}) = (20, 0.5)$ , varying  $\Delta^{\text{prj}} = \lambda^{\text{ob}} - \lambda^{\text{tr}}$  — the same sweep as Fig. 1 ( $b_{\text{sel}}$ -sigmoid) and Fig. 2 ( $\langle \Sigma^{\text{prj}} \rangle$ ). The shape is the familiar two-halo lensing bump peaking around  $\sim R_\lambda$  (dotted  $R_{\text{excl}}$  marker); amplitude scales with  $\Delta^{\text{prj}}$  because both  $b_{\text{sel}}$  plateaus ( $b_{\text{sel}}^{\text{small}}, b_{\text{sel}}^{\text{large}}$ ) grow with projection contamination through Steps 4–5 of the bias pipeline.

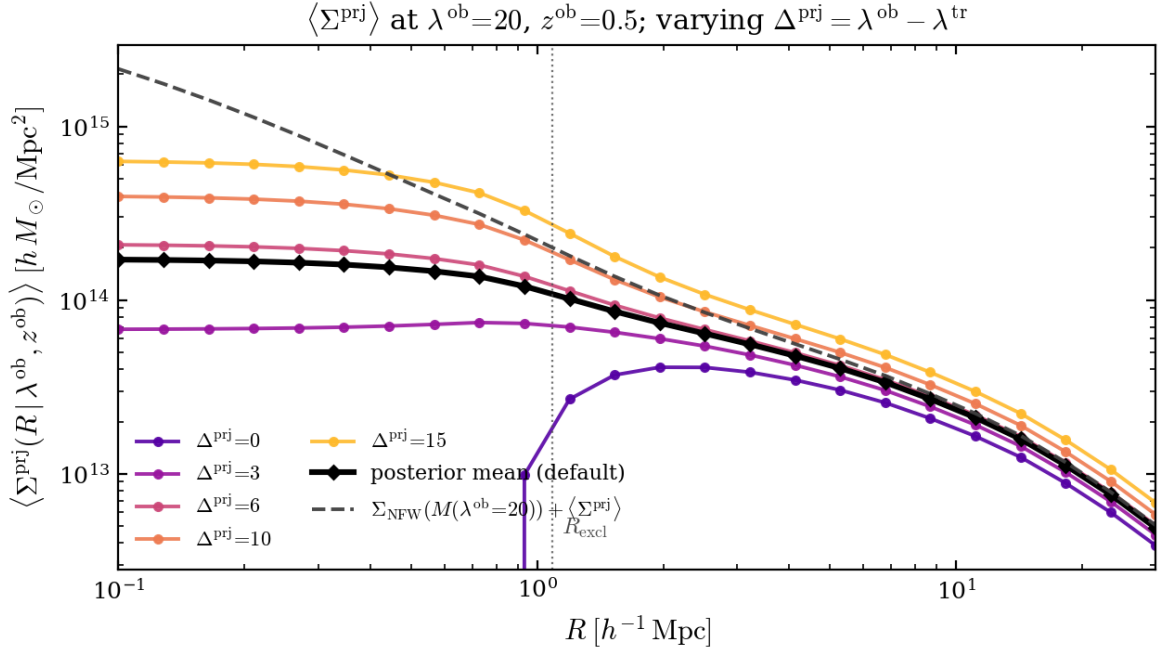


Figure 2: Two-halo  $\langle \Sigma^{\text{prj}}(R | \lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}}) \rangle$  at  $(\lambda^{\text{ob}}, z^{\text{ob}}) = (20, 0.5)$ , varying the projection contamination  $\Delta^{\text{prj}} \equiv \lambda^{\text{ob}} - \lambda^{\text{tr}}$  (same colour palette as Fig. 1).  $b_{\text{sel}}$  is evaluated at fixed  $\lambda^{\text{tr}}$  (not  $\lambda^{\text{tr}}$ -marginalised) to match the Fig. 1 exercise. Dotted line:  $R_{\text{excl}} = R_{\lambda}(\lambda^{\text{ob}})(1 + z^{\text{ob}})$ , separating the miscentered-NFW-dominated inner regime from the correlated  $b_{\text{sel}} \cdot \xi_{\text{NL}}$  large-scale regime.  $\Delta^{\text{prj}} = 0$  sits anomalously low at small  $R$  because both  $b_{\text{sel}}$  plateaus collapse to the halo-bias aggregate in that limit.

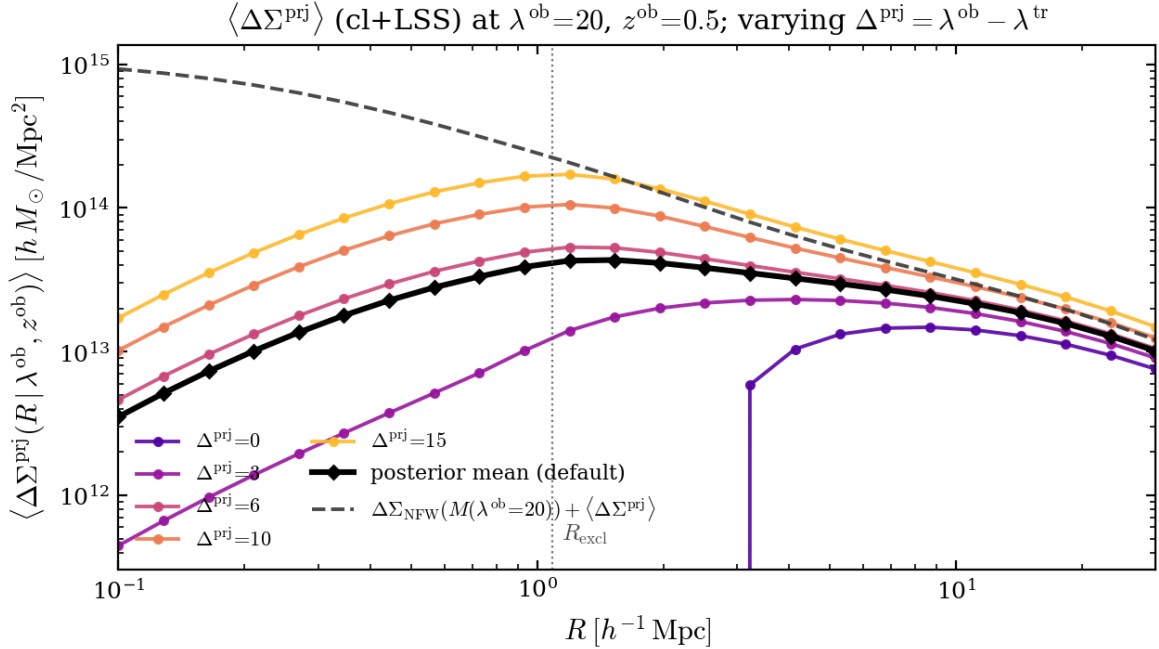


Figure 3:  $\langle \Delta \Sigma^{\text{prj}}(R | \lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}}) \rangle$  at  $(\lambda^{\text{ob}}, z^{\text{ob}}) = (20, 0.5)$ , varying  $\Delta^{\text{prj}} = \lambda^{\text{ob}} - \lambda^{\text{tr}}$  (c1+LSS correlation-excess piece, the class default). Dotted line:  $R_{\text{excl}} = R_{\lambda}(\lambda^{\text{ob}})(1 + z^{\text{ob}})$ . Inner cut-off from the LoS-slab exclusion; outer decay set by the joint support of  $\Delta \Sigma_{\text{mis}}(R | R_{\text{mis}} \sim R)$  and  $\xi_{\text{NL}}(r)$ .

**Precision vs scipy.quad.** Figure 4 compares the production `DeltaSigmaPrj` ( $n_{\text{per\_seg}} = 30$ ,  $N_M = 24$ ,  $N_z = 80$ ,  $R_{\text{max}} = 30 h^{-1} \text{Mpc}$ ) against a nested-`scipy.quad` reference

built by `validations/delta_sigma_prj_diagnostics.py` at the same  $(\lambda^{\text{ob}}, z^{\text{ob}}) = (20, 0.5)$  and  $R \in \{0.3, 1, 3, 10\} h^{-1} \text{ Mpc}$ . The reference is parallelised across the 8 ( $R, \text{kind}$ ) jobs via `multiprocessing.Pool` (8-core wall clock  $\sim 3$  min vs  $\sim 35$  min serial); quad tolerances  $\epsilon_\theta = 3 \times 10^{-3}$ ,  $\epsilon_z = 10^{-2}$ . The residual panel shows sub-0.5% agreement on  $R \geq 1 h^{-1} \text{ Mpc}$  on both **total** and **cl**. At  $R=0.3$  the residual reaches  $\sim 2\%$ : the  $[1 + b_{\text{sel}} \xi_{\text{NL}}]$  bracket changes amplitude sharply where  $\theta_{\text{excl}}(z)$  and  $\theta_R = R/D_A(z^{\text{ob}})$  coincide, and `scipy.quad`'s adaptive refinement issues an `IntegrationWarning` at that crossing. Tightening the quad tolerances by  $3\times$  shifts the reference by  $< 0.01\%$  (so the quad itself is converged), and the GL-on-segments production side is self-converged to  $< 0.5\%$  between  $n_{\text{per\_seg}} = 30$  and  $120$ : the residual at  $R=0.3$  is therefore a real quadrature-method difference at the smallest  $R$ , not a bug in either side. The regression test (`tests/test_delta_sigma_prj.py`) encodes this as a per- $R$  tolerance (0.5% on  $R \geq 1$ , 2.5% at  $R=0.3$ ).

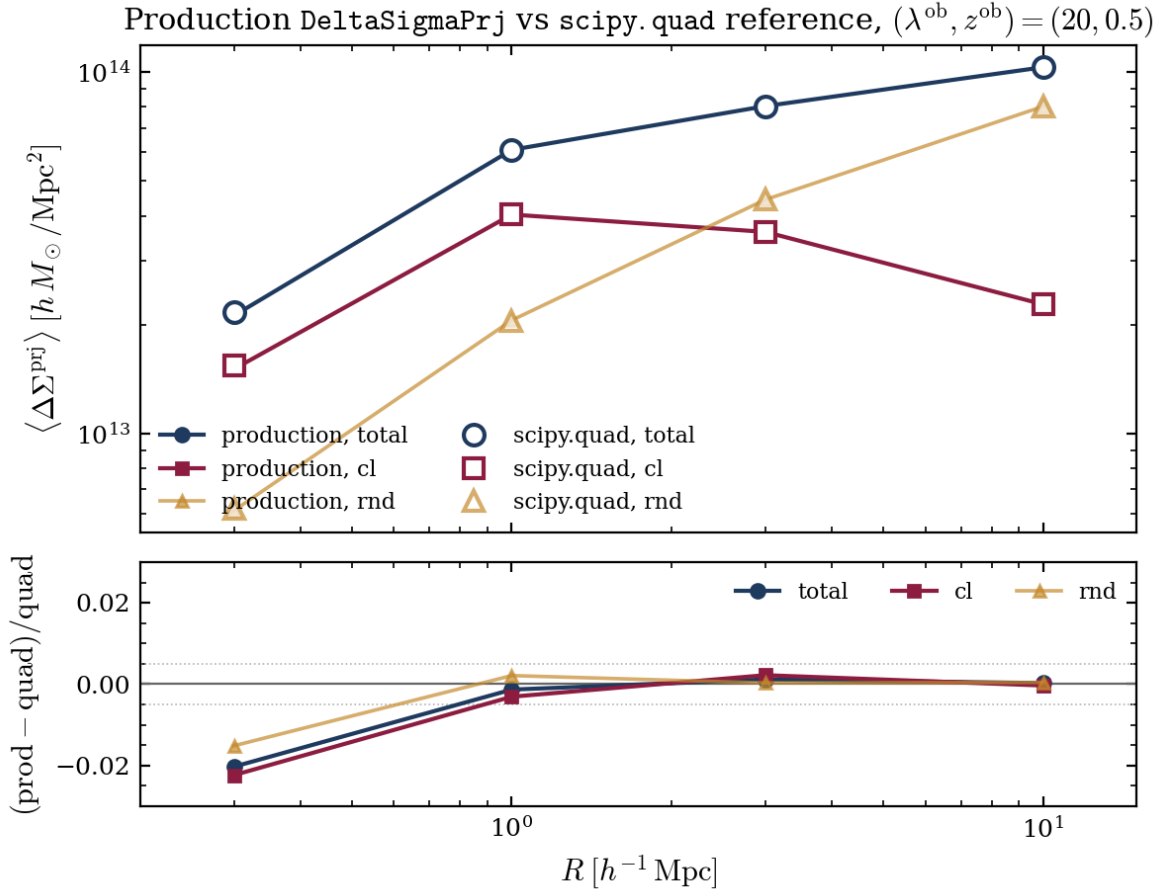


Figure 4: Precision audit of production `DeltaSigmaPrj` (filled markers, solid) against the `scipy.quad` reference (open markers) at  $(\lambda^{\text{ob}}, z^{\text{ob}}) = (20, 0.5)$ . Top: absolute profiles for **total**, **cl** (default return), and **rnd** (truncation-dependent boundary term). Bottom: fractional residual  $(\text{prod} - \text{quad}) / \text{quad}$ , with  $\pm 0.5\%$  guides (dotted). Agreement is sub-0.5% for  $R \geq 1 h^{-1} \text{ Mpc}$ ; at  $R=0.3$  the  $\sim 2\%$  residual tracks a real quadrature-method difference where  $\theta_{\text{excl}}(z) \sim \theta_R$  (see text).

## Numerical pitfalls

The three forward integrals that carry the whole pipeline —  $\mathcal{P}[1]$ ,  $I_1$ ,  $I_2$  of eq. (6) — are 4-D integrals over  $(z, \theta, M, \lambda)$ . The inner  $(M, \lambda)$  axes are well-behaved and a plain Gauss–Legendre (GL) rule with  $N_M = 24$  and  $N_\lambda = 60$  converges them to  $< 10^{-4}$ ; that part of the integration is boring. The outer

$(z, \theta)$  axes, however, contain two sharp, physics-driven features that punish any uninformed fixed-rule method and were the source of the largest precision bugs we saw while developing this package:

- the **halo-exclusion step in  $\theta$** : at each  $z$  the integrand has a hard discontinuity at  $\theta = \theta_{\text{excl}}(z)$ , where  $\xi_{\text{NL}}$  is set to zero by the exclusion prescription. A fixed GL grid on  $(0, 2\theta_\lambda)$  with a mask integrates across this step and stalls at  $\sim 0.3\%$  error regardless of  $N_\theta$  (§3);
- the **exclusion-peak + ring-plateau structure in  $z$** : the photo- $z$  kernel carves out a window of width  $\sim 0.1$  around  $z^{\text{ob}}$ , and inside that window  $\xi_{\text{NL}}$  produces twin sharp peaks at  $z = z^{\text{ob}} \pm \delta z_{\text{excl}}$  sitting on top of a narrow plateau — three qualitatively different regions crammed into  $|z - z^{\text{ob}}| \lesssim 5 \times 10^{-4}$  (§4).

The recipes below move the GL nodes onto the physically correct coordinate in each region so that *the sharp features land at grid boundaries*, where GL rules naturally cluster nodes. The result is sub-0.01% precision on all three integrals at  $N_z = 80$ ,  $N_\theta = 10$  vs `scipy.integrate.quad` with matched inner grids. Total cost at  $(\lambda^{\text{ob}}, z^{\text{ob}}) = (20, 0.5)$ :  $\sim 26$  ms per `SelBias.bias_precompute`. If the bullets below already look familiar, the executive-summary version is: split the  $\theta$ -axis at the exclusion boundary (per- $z$  GL on  $[\theta_{\text{excl}}(z), 2\theta_\lambda]$ , no mask); split the  $z$ -axis by physics into a ring band + two outer halves, integrating in  $z$  on the ring and in  $u = \ln |\Delta\chi_\parallel|$  on the outers.

### 3 The $\theta$ -axis: split at exclusion

**What this section fixes.** The  $\theta$ -axis of  $\mathcal{P}[1], I_1, I_2$  runs over the angular separation between the target and each projected halo. Halo exclusion makes this a discontinuous integrand, which breaks any fixed GL rule that sees the discontinuity as a noisy feature. Below we show why that happens and how moving the lower integration limit onto the discontinuity turns the problem into a smooth integral whose GL convergence is at its design rate.

At any  $z$ , halo exclusion kills  $\theta$  below a threshold  $\theta_{\text{excl}}(z)$  obtained from  $\Delta\chi(z, \theta_{\text{excl}}) = R_{\text{excl}}$ :

$$\cos \theta_{\text{excl}}(z) = \frac{\chi(z)^2 + \chi(z^{\text{ob}})^2 - R_{\text{excl}}^2}{2\chi(z)\chi(z^{\text{ob}})}, \quad \text{clipped to } [-1, 1]. \quad (17)$$

A fixed-grid  $N_\theta=20$  GL on  $(0, 2\theta_\lambda)$  with  $\xi_{\text{NL}} \rightarrow 0$  inside the step integrates across a discontinuity: the error oscillates in  $N_\theta$  and settles to a  $\sim 0.3\%$  floor. The fix is a per- $z$  change of the integration interval: when  $\theta_{\text{excl}}(z) > 0$ , integrate on  $[\theta_{\text{excl}}(z), 2\theta_\lambda]$  with no mask. GL nodes cluster at the endpoints where the step sits; the integrand is smooth on the interval.

| $N_\theta$ | $\mathcal{P}[1]$ err | $I_2$ err      | $I_1$ err      |
|------------|----------------------|----------------|----------------|
| 5          | +0.060%              | +0.046%        | +0.046%        |
| <b>10</b>  | <b>−0.005%</b>       | <b>−0.001%</b> | <b>−0.004%</b> |
| 15         | +0.000%              | +0.000%        | +0.000%        |

$N_\theta = 10$  reaches sub-0.01%:  $\sim 12\times$  node reduction and two decades better precision than the mask recipe.

### 4 The $z$ -axis: split by physics

**What this section fixes.** With  $\theta$  handled, the outer  $z$ -integral becomes the dominant source of residual error — not because it is hard in absolute terms, but because its integrand has three different

shapes over a  $\sim 10^{-3}$ -wide window around  $z^{\text{ob}}$ . A single uniform  $z$ -grid under-samples the exclusion peaks; a single log-spaced grid in  $|\Delta\chi_{\parallel}|$  over-concentrates on the inner edge and misses the ring plateau. The fix is to integrate each physical region on the coordinate in which *that* region is smooth, and stitch them together.

Each of  $\mathcal{P}[1], I_1, I_2$  factorises into an outer  $z$ -integral times an *inner*  $(M, \lambda, \theta)$ -integral,

$$I = \int dz f_X(z), \quad f_X(z) \equiv \frac{dV}{dz d\Omega}(z) w_z(z, z^{\text{ob}}) \mathcal{I}_{\text{inner}}^X(z), \quad (18)$$

where the inner integral is the full  $(M, \lambda, \theta)$  block of eq. (6) evaluated at a single  $z$ ,

$$\mathcal{I}_{\text{inner}}^X(z) \equiv \int dM n(M, z) \int d\lambda P(\lambda | M, z) \lambda \mathbf{1}_{\{\lambda < \lambda^{\text{ob}}\}} \cdot 2\pi \int d\theta \sin\theta f_A(\theta, \lambda^{\text{ob}}, z^{\text{ob}}, \lambda, z) X(M, z, \theta | z^{\text{ob}}), \quad (19)$$

and  $X$  is the per-operator integrand label:  $X = 1$  for  $\mathcal{P}[1]$ ,  $X = b(M, z) \xi_{\text{NL}}(z, z^{\text{ob}}, \theta)$  for  $I_2$ , and  $X = b(M, z) \xi_{\text{NL}}(z, z^{\text{ob}}, \theta) \sigma(\theta)$  for  $I_1$ .  $f_X(z)$  is therefore the  $z$ -density that the outer integral has to quadrature.

For  $\mathcal{P}[1]$ ,  $X = 1$  contains no  $\xi_{\text{NL}}$  and no exclusion;  $f_{\mathcal{P}[1]}(z)$  is a smooth single-peaked function of width  $\sim \sigma_z(z^{\text{ob}})$  peaked at  $z = z^{\text{ob}}$  (Fig. 5 left panel, dashed blue curve). For  $I_1, I_2$  the halo-exclusion cut carves three regions. Defining  $\delta z_{\text{excl}} \equiv R_{\text{excl}}/(c/H(z^{\text{ob}}))$  (at the reference point,  $\approx 4.7 \times 10^{-4}$ ):

1. *Ring plateau*  $|z - z^{\text{ob}}| < \delta z_{\text{excl}}$ : the small- $\theta$  part is excluded; only  $\theta > R_{\text{excl}}/\chi(z^{\text{ob}})$  contributes through a ring of structure at  $\Delta\chi > R_{\text{excl}}$ . Visible as a flat core of  $f_{I_2}(z)$  between the dotted lines in Fig. 5 right panel.
2. *Exclusion peaks* at  $|z - z^{\text{ob}}| \approx \delta z_{\text{excl}}$ :  $\xi_{\text{NL}}$  is evaluated at  $\Delta\chi \rightarrow R_{\text{excl}}$ , its largest resolved value.  $f_{I_2}(z)$  has two sharp peaks at  $z = z^{\text{ob}} \pm \delta z_{\text{excl}}$ , a few times the plateau amplitude (Fig. 5 right panel).
3. *Outer decay*:  $f(z) \propto \Delta\chi_{\parallel}^{-1.7}$  modulated by  $w_z$ , truncating at  $|z - z^{\text{ob}}| \sim \sigma_z(z^{\text{ob}}) \approx 0.09$  (Fig. 5 left panel, outer wings).

Write  $I = I_{\text{ring}} + I_{\text{outer}}^{\text{fg}} + I_{\text{outer}}^{\text{bg}}$  and integrate each region on a grid tailored to its shape. The ring band  $(z^{\text{ob}} - \delta z_{\text{excl}}, z^{\text{ob}} + \delta z_{\text{excl}})$  uses GL in  $z$  (smooth integrand,  $N_{\text{ring}} = \max(9, N_z/4)$ ). The outer pieces use GL in  $u = \ln |\Delta\chi_{\parallel}|$ , so  $\xi_{\text{NL}} \propto e^{-1.7u}$  peaks at the inner endpoint  $u = \ln R_{\text{excl}}$  where GL naturally clusters nodes. Per side:  $N_{\text{outer}} = \max(15, (N_z - N_{\text{ring}})/2)$ . Each region's integrand is smooth in its own variable; the sharp features sit at region boundaries where GL rules converge at their design rate.

At  $N_z = 80$  this reaches sub-0.01% on  $\mathcal{P}[1], I_1, I_2$  vs `scipy.integrate.quad` with matched inner grids (regression test `tests/test_integrals.py::TestSelBiasThetaSplit`). Edge cases ( $R_{\text{excl}}$  exceeds the photo- $z$  support, or the ring band is wider than it) degrade gracefully via empty-array guards.

$z$ -integrand at  $\lambda^{\text{ob}}=20$ ,  $z^{\text{ob}}=0.5$  ( $\delta z_{\text{excl}} \approx 4.7e-04$ )

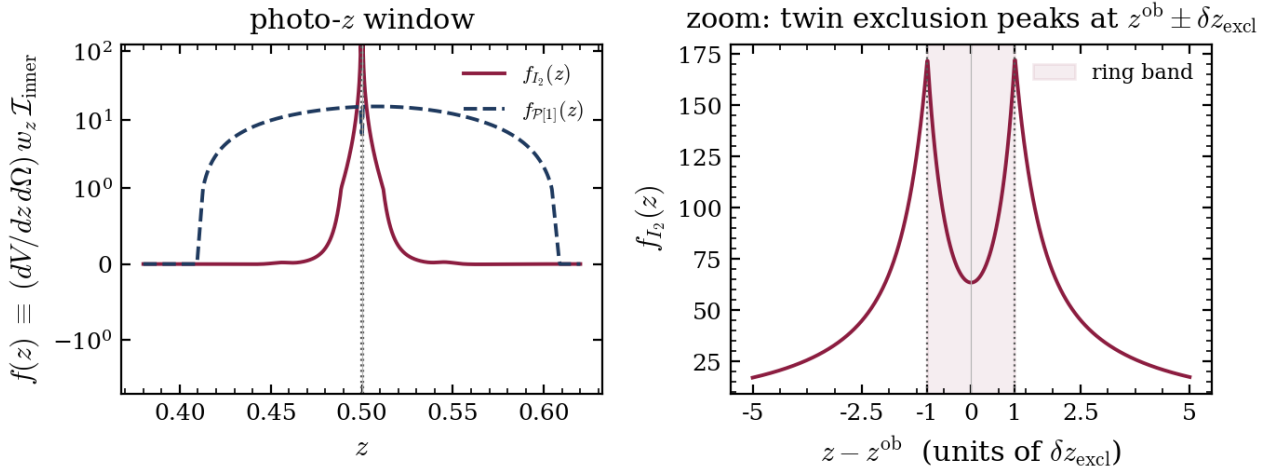


Figure 5: The  $z$ -integrand  $f_X(z)$  of eq. (18) at  $(\lambda^{\text{ob}}, z^{\text{ob}}) = (20, 0.5)$ . *Left*: full photo- $z$  window, symlog  $y$ -scale; the red solid curve is  $f_{I_2}(z)$ , the blue dashed curve is  $f_{\mathcal{P}[1]}(z)$ .  $\mathcal{P}[1]$  (no  $\xi_{\text{NL}}$ ) is smooth and single-peaked;  $I_2$  has an additional two-peak exclusion substructure at  $z^{\text{ob}} \pm \delta z_{\text{excl}}$  (dotted vertical lines) that is invisible at this scale but inflated by the symlog axis. *Right*: zoom to  $|z - z^{\text{ob}}| < 5 \delta z_{\text{excl}}$ , linear scale. The twin exclusion peaks are resolved; the ring-plateau band between them (red shading) is where  $\xi_{\text{NL}}$  is excluded and the integrand is flat. A naive uniform  $z$ -grid places zero nodes inside this window and under-samples both peaks, which is the hazard the split-by-physics recipe avoids.

## API and glossary

This section is a pointer table from the math above to the code that computes each quantity. The glossary table lists every symbol with its defining equation, a one-line description, and the Python call that returns it.

### 5 Objects

**Cosmology** (`cosmology.py`). Flat- $\Lambda$ CDM background. Provides `chi(z) =  $\chi(z)$`  comoving distance, `dV_dzdOm(z) =  $dV/dz d\Omega$` , and `h, H0`. Default: Buzzard-v1.1.

**PkGrid** (`pk.py`). CAMB-backed linear  $P(k, z)$ . One CAMB call per cosmology; re-evaluated in place for each  $(k, z)$  request.

**SigmaM** (`sigma_m.py`).  $\sigma(M, z)$  top-hat variance via a Simpson rule on PkGrid’s  $k$ -grid. Used by both HMF and Bias.

**HMF** (`hmf.py`). Tinker 2008,  $\Delta = 200$ . `hmf(M, z)` returns  $n(M, z)$  in  $h^{-1} M_{\odot}^{-1} h^{-1} \text{Mpc}^{-3}$ .

**Bias** (`bias.py`). Tinker 2010 peak-height halo bias. `bias(M, z)` returns  $b(M, z)$ .

**MOR** (`mor.py`). Mass–observable relation. `mor.pdf(lam, M, z)` returns  $P(\lambda | M, z)$ , the EMG kernel of Costanzi 2019.

**XiNL** (`xi_nl.py`). Nonlinear matter correlation function  $\xi_{\text{NL}}(r, z)$  from halofit  $P_{\text{NL}}$  via an FFTlog Hankel transform. Lazy build (first call pays  $\sim 2$  s, cached thereafter).

**SelBias** (`sel_bias.py`). The bias pipeline (§2.4). Holds references to all of the above. See the glossary below for per-method meanings.

**NFWMiscentered** (`nfw.py`).  $\Sigma_{\text{mis}}(R | M, z, \theta, z^{\text{ob}})$  from the three offset-NFW lookup tables (Costanzi-notebook convention).

**SigmaPrj** (`sigma_prj.py`). The  $\langle \Sigma^{\text{prj}} \rangle$  orchestrator (eq. 14). `SigmaPrj(cosmo, sel_bias, nfw)(R, lob, zob)` returns  $\langle \Sigma^{\text{prj}}(R) \rangle$ .

**GridConfig** (`config.py`). Grid-size defaults:  $N_z = 80$ ,  $N_{\theta} = 10$ ,  $N_M = 24$ .  $N_{\lambda}$  is a separate **SelBias** constructor arg (default 60).

### 6 Glossary of quantities

Rows group by role: halo ingredients, projection ingredients, the  $\mathcal{P}$  operator specialisations, the bias-pipeline outputs, and the profile. “Eq.” links to the defining equation; “Code” gives the Python call.

| Symbol                                                                               | Meaning                                                                                                          | Eq.  | Code                                                        |
|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|------|-------------------------------------------------------------|
| <i>Halo ingredients</i>                                                              |                                                                                                                  |      |                                                             |
| $n(M, z)$                                                                            | Tinker-2008 halo mass function ( $\Delta = 200$ )                                                                | (3)  | <code>HMF(sigma_m)(M, z)</code>                             |
| $b(M, z)$                                                                            | Tinker-2010 peak-height halo bias                                                                                | (3)  | <code>Bias(sigma_m)(M, z)</code>                            |
| $P(\lambda   M, z)$                                                                  | MOR kernel (Costanzi-2019 EMG)                                                                                   | (6)  | <code>MOR().pdf(lam, M, z)</code>                           |
| $\xi_{\text{NL}}(r, z)$                                                              | Nonlinear matter 2pt correlation (halofit+FFTlog); zero below $R_{\text{excl}}$                                  | (6)  | <code>XiNL(cosmo)(r, z)</code>                              |
| <i>Projection kernel and <math>\mathcal{P}[X]</math> operator</i>                    |                                                                                                                  |      |                                                             |
| $\rho^{\text{prj}}$                                                                  | Per-halo projection weight $w_z f_A \lambda \mathbf{1}_{\{\lambda < \lambda^{\text{ob}}\}}$                      | (5)  | internal to <code>SelBias.P_operator</code>                 |
| $\mathcal{P}[X]$                                                                     | LoS average of $X$ weighted by $\rho^{\text{prj}}$ and halo distribution                                         | (6)  | <code>SelBias.P_operator(lob, zob)</code>                   |
| $\mathcal{P}[1] \equiv \langle \Delta_{\text{bkg}}^{\text{prj}} \rangle$             | Mean background projection (no $\xi_{\text{NL}}$ , smooth in $z$ )                                               | (6)  | <code>bias_precompute(lob,zob)['P1']</code>                 |
| $I_2 \equiv \mathcal{P}[b \xi_{\text{NL}}]$                                          | Halo-bias-weighted $\xi_{\text{NL}}$ ; drives $\Delta_{\text{RND}}^{\text{prj}}$                                 | (6)  | <code>bias_precompute(lob,zob)['I2']</code>                 |
| $I_1 \equiv \mathcal{P}[b \xi_{\text{NL}} \sigma(\theta)]$                           | $I_2$ weighted by the sigmoid; enters Step 5                                                                     | (6)  | <code>bias_precompute(lob,zob)['I1']</code>                 |
| <i>Bias-pipeline outputs</i>                                                         |                                                                                                                  |      |                                                             |
| $b_{\text{halo}}(\lambda^{\text{ob}}, z^{\text{ob}})$                                | Halo-bias aggregate at fixed $\lambda^{\text{ob}}$ (paper's $b_{\text{eff}}$ )                                   | (3)  | <code>SelBias.b_eff(lob, zob)</code>                        |
| $\Delta_{\text{RND}}^{\text{prj}}$                                                   | Projection contamination of a random aperture                                                                    | (8)  | <code>bias_precompute(lob,zob)['Delta_RND']</code>          |
| $b_{\text{sel}}^{\text{large}}$                                                      | Large- $\theta$ plateau (empirical closure)                                                                      | (9)  | <code>bias_from_precomp(pre,ltr)['b_infty']</code>          |
| $b_{\text{sel}}^{\text{small}}$                                                      | Small- $\theta$ plateau (Step 5 inversion)                                                                       | (10) | <code>bias_from_precomp(pre,ltr)['b_zero']</code>           |
| $b_{\text{sel}}(\lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}}, \theta)$    | Scale-dependent bias via sigmoid ansatz                                                                          | (4)  | <code>SelBias.b_lob_theta(th,ltr,zob,lob,precomp)</code>    |
| $\bar{b}_{\text{small}}, \bar{b}_{\text{large}}$                                     | $\lambda^{\text{tr}}$ -averaged plateaus (cached per $(\lambda^{\text{ob}}, z^{\text{ob}})$ )                    | (12) | <code>SelBias.marginalised_plateaus(lob,zob,precomp)</code> |
| $b_{\text{sel}}(\lambda^{\text{ob}}, z^{\text{ob}}, \theta)$                         | $\lambda^{\text{tr}}$ -marginalised bias; input to $\langle \Sigma^{\text{prj}} \rangle$                         | (13) | <code>SelBias.b_sel_marginalised(th,lob,zob,precomp)</code> |
| <i>Two-halo profile</i>                                                              |                                                                                                                  |      |                                                             |
| $\Sigma_{\text{mis}}(R   M, z, \theta, z^{\text{ob}})$                               | Azimuth-averaged miscentered NFW (offset-NFW lookup)                                                             | (15) | <code>NFWMiscentered(cosmo).sigma_grid(R,Rtheta)</code>     |
| $\Delta \Sigma_{\text{mis}}$                                                         | Miscentered NFW radial excess $\bar{\Sigma}_{\text{mis}}(< R) - \Sigma_{\text{mis}}(R)$ (companion lookup table) | —    | <code>NFWMiscentered.delta_sigma_grid</code>                |
| $\langle \Sigma^{\text{prj}}(R   \lambda^{\text{ob}}, z^{\text{ob}}) \rangle$        | Two-halo projected surface density (main output)                                                                 | (14) | <code>SigmaPrj(cosmo,sel_bias,nfw)(R,lob,zob)</code>        |
| $\langle \Delta \Sigma^{\text{prj}}(R   \lambda^{\text{ob}}, z^{\text{ob}}) \rangle$ | Two-halo lensing excess (see docs/delta_sigma_prj_derivation.tex)                                                | —    | <code>DeltaSigmaPrj(cosmo,sel_bias,nfw)(R,lob,zob)</code>   |

## 7 Validation

**Unit tests.** `tests/test_integrals.py`: `TestSelBiasThetaSplit` (sub-0.1% vs quad at  $N_\theta = 10$ ), `TestLambdaAxisConverged` ( $n_\lambda^{\text{tr}} = 30$  matches 100 to  $< 10^{-4}$ ), `TestSigmaPrjTheta` (split-at- $\theta_R +$

monotonicity). `test_smoke.py`: end-to-end smoke test.

**Validation scripts.** `validations/`: `theta_convergence.py` sweeps  $N_\theta$ ; `quad_matched_inner.py` compares the `_P_operator` output to `scipy.integrate.quad` with matched inner GL grids; `quad_validate.py` is the end-to-end quad cross-check.

**Defaults.**  $N_z = 80$ ,  $N_\theta = 10$ ,  $N_M = 24$ ,  $N_\lambda = 60$ . No retuning is needed across the Y3 range ( $\lambda^{\text{ob}} \in [20, 500]$ ,  $z^{\text{ob}} \in [0.1, 0.8]$ ): only  $R_{\text{excl}}$  and  $c/H(z^{\text{ob}})$  shift, and both are tracked analytically by the grid builder.

## References

Costanzi, M., Wu, H.-Y., Esteves, J. H., Grandis, S., To, C., & Aguena, M. 2026, Phys. Rev. D (accepted), arXiv:2604.05833.